

# Dimitrios Panagiotis Emmanouilidis

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## RESEARCH INTERESTS

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Measurement and evaluation of LLM-driven control systems; local inference optimisation; multi-agent architectures; Retrieval-Augmented Generation.

## EDUCATION

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### BSc (Hons) Computing (Application Development)

New York College in partnership with University of Greater Manchester — Thessaloniki, Greece

Expected July 2026

- Level 6 Weighted Average: 62.83% (Upper Second-Class Honours equivalent). Cumulative Average: 53.39%.
- Dissertation: "Contextual Synthesis and Constrained Action: Evaluating a Two-Agent LLM System for Personalized Vehicle Adaptation" (May 2026, 76%).
- Research & Analysis: Undergraduate Research Project (76%), Contemporary Problem Analysis.
- Software Engineering: Introduction to Software Development (76%), Enterprise Systems Development (67%), Data Structures and Algorithms, Object Oriented Programming, Open Source Software Development.
- Systems & Data: Introduction to Networking (73%), Database Design, Secure Application Development, Server Application Configuration and Management.

## RESEARCH & TECHNICAL PROJECTS

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### actuation-runner — LLM Action-Payload Failure Benchmark *Sole Author · Public, Aug 2026*

- Open-source instrument measuring how often LLM-generated action payloads fail before reaching an actuator, across eight mutually exclusive failure categories.
- Isolates out-of-bounds values: payloads that are schema-valid and fail only against a physical limit. No structured-output benchmark surveyed measures this separately.
- Two 100-trial runs on a local quantised 9B model: 89/100 and 92/100 valid, reported separately so run-to-run variance stays visible.
- Every failure across 200 trials was planner token-ceiling truncation, with termination reason matching outcome one-to-one. Upstream truncation occurred 7 times and never caused failure.
- 45-test offline suite exercising the real invocation path; caught three defects before the measured run. Apache 2.0.

### Two-Agent LLM Pipeline with Deterministic Validation Gate *Undergraduate Dissertation · Early 2026*

- Two-stage LLM pipeline, profiler feeding planner, behind a non-LLM JSON Schema gate, tested in CARLA under synchronous and asynchronous execution.
- 4-bit quantised 9B model via Ollama in a 12 GB VRAM budget shared with the simulator.
- Measured end-to-end latency and reported it as prohibitive for real-time control on that hardware.
- Audited my own instrumentation afterwards and established that a reported failure-rate decomposition could not have been produced by the logging implemented. That finding motivated actuation-runner.
- Isolated the dominant latency mechanism as token-ceiling truncation rather than hardware limitation, and found an idle simulator halved throughput with GPU utilisation unchanged.

### Offline Local Inference & Retrieval Application *Lead Developer · In Development*

- Local LLM application for off-grid retrieval, with no cloud API dependency.
- Quantised models tuned for consumer hardware across mobile and eGPU setups.
- RAG pipeline over large offline databases such as Wikipedia and medical guides.

### Local AI Infrastructure & Orchestration *Independent Work · Late 2025 – Present*

- Local AI execution environment: Beelink GTi13 Ultra with external NVIDIA RTX 5070.
- Deployed LangGraph and Ollama, tuning VRAM allocation and inference performance.

## TECHNICAL SKILLS

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**Languages** Python

**AI & Machine Learning** Ollama, LangGraph, local LLM inference, quantised models, MCP, RAG, multi-agent orchestration, JSON Schema validation, benchmark design

**Infrastructure** Docker, Git, VRAM management, eGPU configuration, MLOps

## CERTIFICATIONS

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- Python Development Certification (LearnX) — March 2025

## LANGUAGES

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Greek: Native | English: C2 Level Certification